

exercises

July 15, 2026

1 Infinite Series, Power Series

```
[1]: %load_ext autoreload
      %autoreload 2

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import math
from IPython import display
from IPython.display import Markdown
from scipy.special import factorial
from scipy import stats
from scipy import constants

# Below are some adjustments:
# from IPython.display import HTML
# HTML('<style>img[alt="skan"] { filter: invert(100%); }</style>')
# plt.style.use('dark_background')
```

1.1 Problems, Section 2

In the following problems, find the limit of the given sequence as $n \rightarrow \infty$.

1.1.1 2.6

$$\lim_{n \rightarrow \infty} \frac{n^n}{n!}$$

```
[2]: # 1. Simulation setup
      # We keep the range small (1 to 10) because  $n^n$  grows insanely fast.
      n_max: float = 10.0
      step: float = 0.1
      start: float = 1.0
      n_values = np.arange(start, n_max, step)
      custom_mk = 10

      numerators = n_values**n_values
```

```

denominators = factorial(n_values)
results = numerators / denominators

# 3. Plotting the chart
plt.figure(figsize=(10, 6))

# Draw the main line
plt.plot(n_values, results, color='red', marker='o', markevery=custom_mk,
        label=r'values  $\frac{n^n}{n!}$ ')

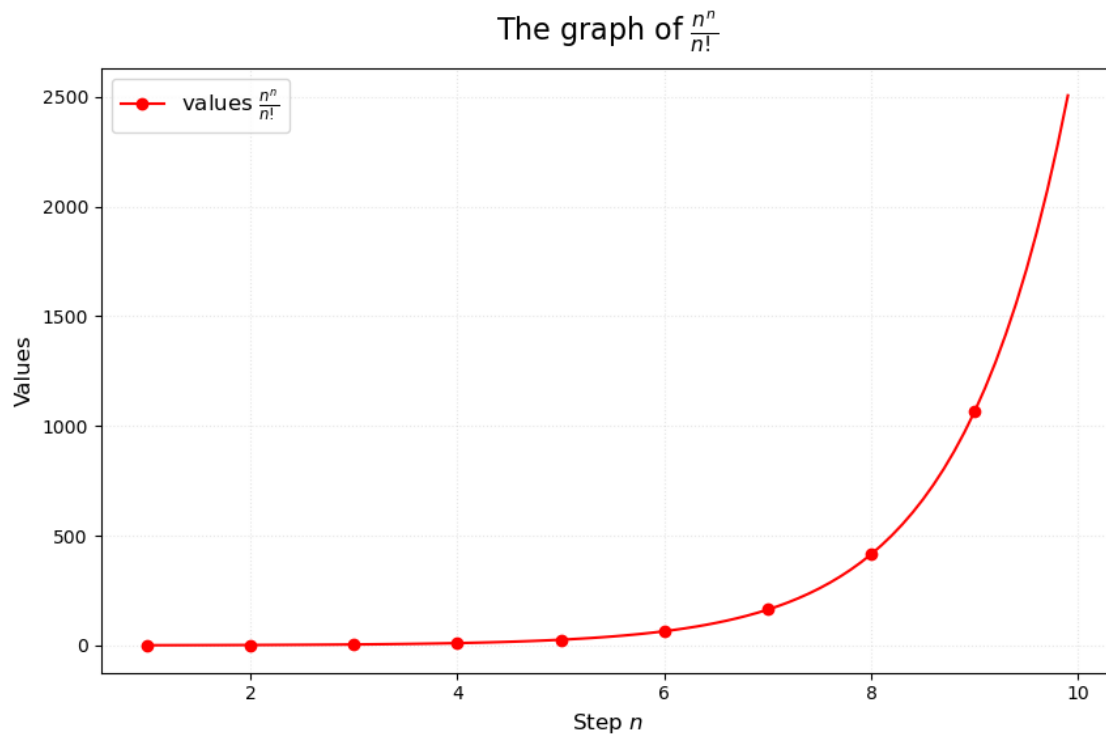
# You may ask "why is this code excluded?"
# Let's have a look at the table of Values below
# It's reasonable not to include numerator and denominator because of the rapid
    growth of those values
#plt.plot(n_values, numerators, color='blue', linestyle='-', marker='s',
    markevery=5, label=r'numerator  $n^n$ ')
#plt.plot(n_values, denominators, color='green', linestyle='--', marker='^',
    markevery=5, label=r'denominator  $n!$ ')

# Aesthetics and labels
plt.title(r'The graph of  $\frac{n^n}{n!}$ ', fontsize=16, pad=15)
plt.xlabel('Step  $n$ ', fontsize=12)
plt.ylabel('Values', fontsize=12)
plt.grid(True, linestyle=':', alpha=0.3)
plt.legend(fontsize=12)

# Render the plot
plt.show()

table_data = {
    r"Step  $n$ ": n_values,
    r"Numerator  $n^n$ ": numerators,
    r"denominator  $n!$ ": denominators,
    r"Value  $\frac{n^n}{n!}$ ": results
}
df = pd.DataFrame(table_data)
df_sliced = df.iloc[:10]
Markdown(df_sliced.to_markdown(index=False))

```



[2] :

Step n	Numerator n^n	denominator $n!$	Value $\frac{n^n}{n!}$
1	1	1	1
2	4	2	2
3	27	6	4.5
4	256	24	10.6667
5	3125	120	26.0417
6	46656	720	64.8
7	823543	5040	163.401
8	1.67772e+07	40320	416.102
9	3.8742e+08	362880	1067.63

Using our simulation, it is reasonable to say that:

$$\lim_{n \rightarrow \infty} \frac{n^n}{n!} = \infty$$

1.1.2 2.7

$$\lim_{n \rightarrow \infty} \left((1 + n^2)^{\frac{1}{\ln n}} \right)$$

Simulation:

```

[3]: def compute():
    n_max = 20
    n_start = 2.0
    n_step = 0.1
    n_values = np.arange(n_start, n_max, n_step)
    results = (1 + n_values**2)**(1 / np.log(n_values))
    powers = 1 / np.log(n_values)
    bases = 1 + n_values**n_values

    plt.figure(figsize=(10,6))

    plt.plot(n_values, results, color='red', linestyle="--", marker='o',
    ↪markevery=10, label=r'Values  $(1+n^2)^{\frac{1}{\ln n}}$ ')

    euler_value = np.exp(2)

    plt.axhline(y=euler_value, color='blue', linestyle='--',linewidth=2,
    ↪label=r'$e^2$')
    plt.title(r"The graph of  $(1+n^2)^{\frac{1}{\ln n}}$ ", fontsize=16, pad=15)
    plt.xlabel('$n$', fontsize=14)
    plt.ylabel('Values', fontsize=14)
    plt.grid(True, linestyle=':', alpha=0.5)
    plt.legend(fontsize=14, loc='upper right')

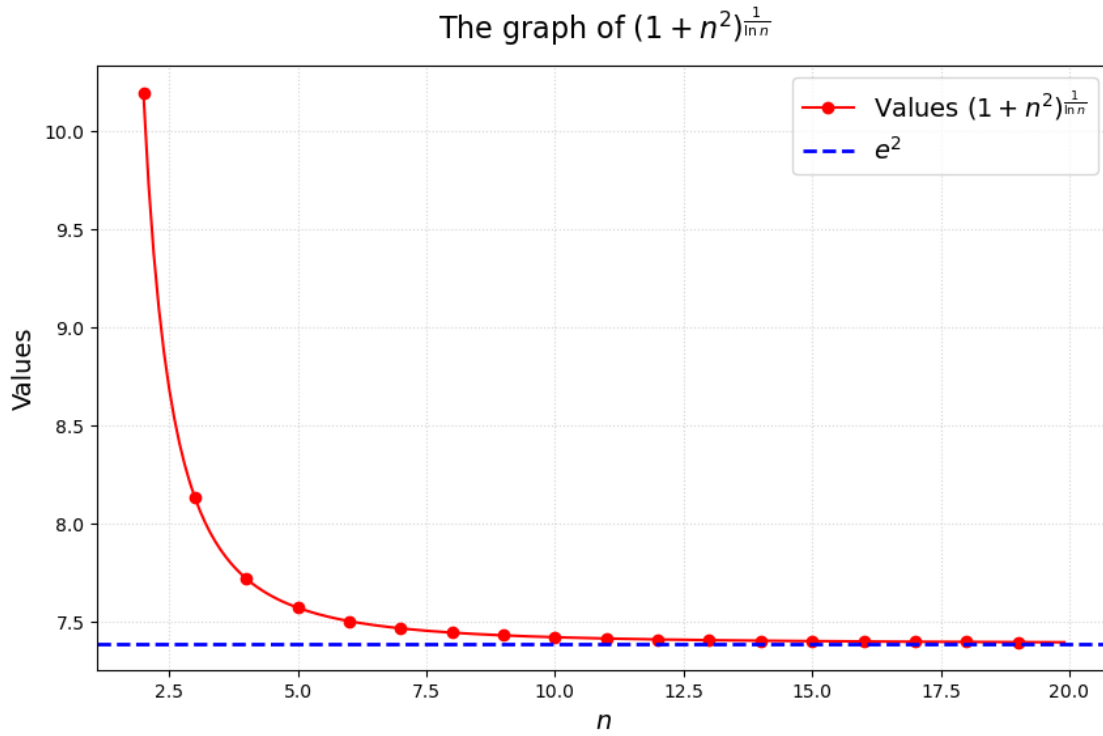
    plt.show()

    table_data = {
        r"Step $n$": n_values,
        r"Base $1 + n^2$": bases,
        r"Power  $\frac{1}{\ln n}$ ": powers,
        r"Values  $(1+n^2)^{\frac{1}{\ln n}}$ ": np.round(results, 4)
    }

    df = pd.DataFrame(table_data)
    df_sliced = df.iloc[:20]
    return Markdown(df_sliced.to_markdown(index=False))

compute()

```



[3] :

Step n	Base $1 + n^2$	Power $\frac{1}{\ln n}$	Values $(1 + n^2)^{\frac{1}{\ln n}}$
2	5	1.4427	10.1953
4	257	0.721348	7.7194
6	46657	0.558111	7.5029
8	1.67772e+07	0.480898	7.4444
10	1e+10	0.434294	7.4211
12	8.9161e+12	0.40243	7.4097
14	1.1112e+16	0.378923	7.4033
16	1.84467e+19	0.360674	7.3995
18	3.93464e+22	0.345976	7.3969

1.1.3 2.8

$$\lim_{n \rightarrow \infty} \frac{(n!)^2}{(2n)!}$$

```
[4]: def compute():
    n_max = 10
    n_step = 0.1
    n_start = 2
    n_values = np.arange(n_start, n_max, n_step)
```

```

numerators = factorial(n_values)**2
denominators = factorial(2*n_values)
results = numerators / denominators

plt.figure(figsize=(16,9))

plt.plot(n_values, results, 'o-r', markevery=10, linewidth=3,
↳label=r'Values  $\frac{(n!)^2}{(2n)!}$ ')
    #plt.plot(n_values, numerators, color='green', marker='^', linestyle='-',
↳markevery=11, label=r'Numerators  $(n!)^2$ ')

plt.title(r'the graph of:  $\frac{(n!)^2}{(2n)!}$ ', fontsize=30, pad=15)
plt.xlabel(r"$n$", fontsize=16)
plt.ylabel(r"Values", fontsize=16)
plt.grid(True, linestyle=':', alpha=0.5)
plt.legend(fontsize=16, loc='upper right')

plt.show()

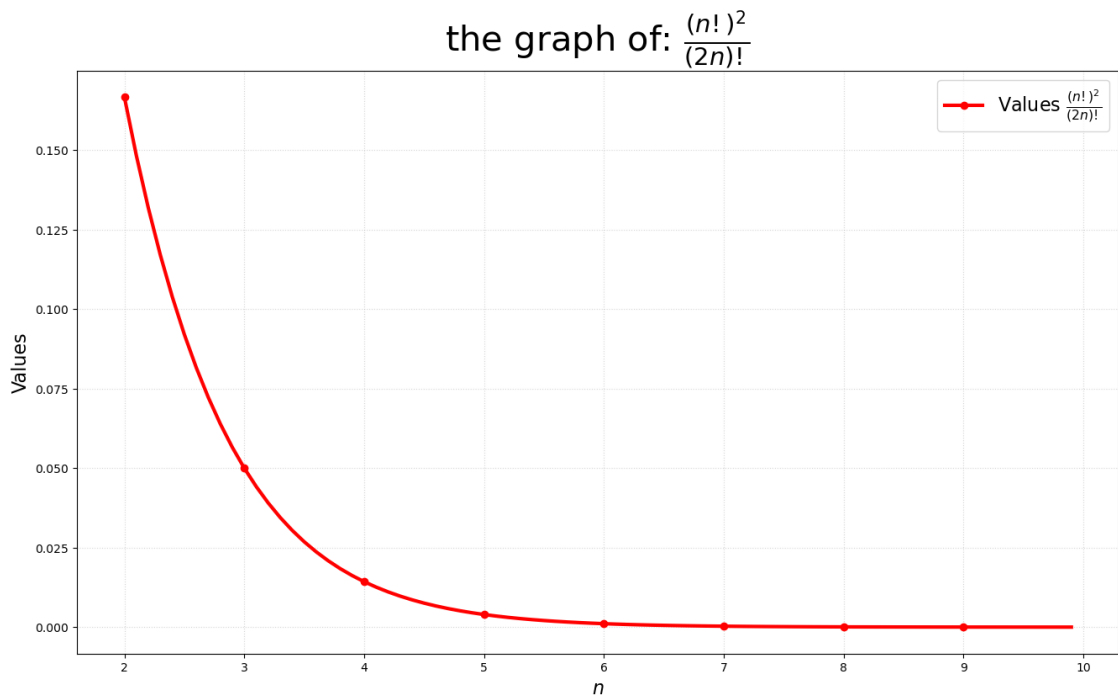
table_data = {
    r'Step $n$': n_values,
    r'Numerator  $(n!)^2$ ': numerators,
    r'Denominator'  $(2n)!$ ': denominators,
    r"Values  $\frac{(n!)^2}{(2n)!}$ ": results
}

df = pd.DataFrame(table_data)
df_sliced = df.iloc[::10]

return Markdown(df_sliced.to_markdown(index=False))

compute()

```



[4] :

Step n	Numerator $(n!)^2$	Denominator' $(2n)!$	Values $\frac{(n!)^2}{(2n)!}$
2	4	24	0.166667
3	36	720	0.05
4	576	40320	0.0142857
5	14400	3.6288e+06	0.00396825
6	518400	4.79002e+08	0.00108225
7	2.54016e+07	8.71783e+10	0.000291375
8	1.6257e+09	2.09228e+13	7.77001e-05
9	1.31682e+11	6.40237e+15	2.05677e-05

2.8

$$\lim_{n \rightarrow \infty} \frac{(n!)^2}{(2n)!}$$

$$(n!)^2 = (1 \cdot 2 \cdot 3 \cdot \dots \cdot n)(1 \cdot 2 \cdot 3 \cdot \dots \cdot n)$$

$$(2n)! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n(n+1) \cdot (n+2) \cdot \dots \cdot 2n$$

$$a_n = \frac{n! \cdot n!}{n! \cdot (n+1)(n+2) \cdots 2n} = \frac{n!}{(n+1)(n+2) \cdots 2n} =$$

$$= \frac{1}{n+1} \cdot \frac{2}{n+2} \cdot \frac{3}{n+3} \cdots \frac{n}{n+n}$$

$$a_n \leq \left(\frac{1}{2}\right)^n \Rightarrow \lim_{n \rightarrow \infty} \frac{(n!)^2}{(2n)!} = 0$$

1.1.4 2.9

$$\lim_{n \rightarrow \infty} n \sin\left(\frac{1}{n}\right)$$

2.9

$$\lim_{n \rightarrow \infty} n \sin \frac{1}{n} = [0, \infty]$$

$\nearrow \infty$
 $\searrow 0$

$$x = \frac{1}{n} \Rightarrow n = \frac{1}{x}$$

As $n \rightarrow \infty$, $\frac{1}{x}$ approaches 0

$$\lim_{n \rightarrow \infty} n \sin \frac{1}{n} = \lim_{x \rightarrow 0} \frac{1}{x} \sin x = \lim_{x \rightarrow 0} \frac{\sin x}{x} =$$

$$= \lim_{n \rightarrow \infty} 1 \cdot \frac{\sin x}{x} = 1$$

1.2 Problems, Section 4

For the following series, write formulas for the sequences a_n , S_n , and R_n , and find the limits of the sequences as $n \rightarrow \infty$ (if the limits exist).

1.2.1 4.1

$$\sum_{n=1}^{\infty} \frac{1}{2^n}$$

4.1.

$$\sum_1^{\infty} \frac{1}{2^n}$$

$$a_n = \frac{1}{2^n}$$

$$S_n = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n}$$

$$\left. \begin{array}{l} a_1 = \frac{1}{2} \\ r = \frac{1}{2} \end{array} \right\} S_n = \frac{a_1 \cdot (1 - r^n)}{1 - r} = \frac{\frac{1}{2} [1 - (\frac{1}{2})^n]}{1 - \frac{1}{2}}$$

$$S_n = \frac{\frac{1}{2} - (\frac{1}{2})^{n+1}}{\frac{1}{2}} = 1 - (\frac{1}{2})^n$$

$$R_n = S - S_n = \lim_{n \rightarrow \infty} S_n - S_n =$$

$$= 1 - 1 + (\frac{1}{2})^n = (\frac{1}{2})^n$$

1.2.2 4.2

$$\sum_0^{\infty} \frac{1}{5^n}$$

h.2

$$\sum_{n=0}^{\infty} \frac{1}{5^n} = 1 + \frac{1}{5} + \frac{1}{25} + \dots$$

I. $a_n = \frac{1}{5^{n-1}}$ *usually, geometric progression starts with $n=1$, but in this case $a_1=1$, so we need to subtract 1 from n*

$$\text{II. } a_1=1, r=\frac{1}{5} \Rightarrow S_n = 1 \cdot \frac{[1 - (\frac{1}{5})^n]}{1 - \frac{1}{5}} = \frac{1 - (\frac{1}{5})^n}{\frac{4}{5}} = \frac{5 - (\frac{1}{5})^{n-1}}{4} = \frac{5}{4} [1 - (\frac{1}{5})^n]$$

$$\text{III. } \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[\frac{5}{4} [1 - (\frac{1}{5})^n] \right] = \left[\text{as } n \rightarrow \infty, (\frac{1}{5})^n \rightarrow 0 \right] = \frac{5}{4}$$

$$\text{IV. } R_n = S - S_n = \frac{5}{4} - \frac{5}{4} [1 - (\frac{1}{5})^n] = \frac{5}{4} - \frac{5}{4} + \frac{5}{4} (\frac{1}{5})^n = \frac{1}{4} \cdot (\frac{1}{5})^{n-1}$$

$$\text{V. } \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{1}{4} \cdot (\frac{1}{5})^{n-1} = \left[\text{as } n \rightarrow \infty, (\frac{1}{5})^{n-1} \rightarrow 0 \right] = 0$$

1.2.3 4.3

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} \dots$$

h.3

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16}$$

$$\text{I. } a_n = \frac{1}{(-2)^{n-1}}$$

$$\text{II. } a_1=1, r=-\frac{1}{2} \Rightarrow S_n = 1 \cdot \frac{1 - (-\frac{1}{2})^n}{1 - (-\frac{1}{2})} = \frac{1 - (-\frac{1}{2})^n}{\frac{3}{2}} = \frac{2}{3} [1 - (-\frac{1}{2})^n]$$

$$\text{III. } \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{2}{3} [1 - (-\frac{1}{2})^n] = \left[\text{as } n \rightarrow \infty, (-\frac{1}{2})^n \rightarrow 0 \right] = \frac{2}{3}$$

$$\text{IV. } R_n = S - S_n = \frac{2}{3} - \frac{2}{3} [1 - (-\frac{1}{2})^n] = \frac{2}{3} (-\frac{1}{2})^n$$

$$\text{V. } \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{2}{3} (-\frac{1}{2})^n = \left[\text{as } n \rightarrow \infty, (-\frac{1}{2})^n \rightarrow 0 \right] = 0$$

1.2.4 4.4

$$\sum_{n=1}^{\infty} e^{-n \ln 3}$$

$$\begin{aligned}
 & \sum_{n=1}^{\infty} e^{-n \ln 3} \quad \nearrow \ln 3 = \log_e 3, \quad x = e^{\ln x} \\
 & e^{-n \ln 3} = (e^{-\ln 3})^n = \left(\frac{1}{3}\right)^n \\
 \text{I} \quad & a_n = \left(\frac{1}{3}\right)^n \\
 \text{II} \quad & \left. \begin{array}{l} a_1 = \frac{1}{3} \\ r = \frac{1}{3} \end{array} \right\} S_n = \frac{1}{3} \cdot \frac{1 - \left(\frac{1}{3}\right)^n}{1 - \frac{1}{3}} = \frac{1}{3} \cdot \frac{2}{2} \cdot [1 - \left(\frac{1}{3}\right)^n] = \frac{1}{2} [1 - \left(\frac{1}{3}\right)^n] \\
 \text{III} \quad & \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{1}{2} [1 - \left(\frac{1}{3}\right)^n] = [as \, n \rightarrow \infty, \left(\frac{1}{3}\right)^n \rightarrow 0] = \frac{1}{2} \\
 \text{IV} \quad & R_n = S - S_n = \frac{1}{2} - \frac{1}{2} [1 - \left(\frac{1}{3}\right)^n] = \frac{1}{2} \left(\frac{1}{3}\right)^n \\
 \text{V} \quad & \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{1}{2} \left(\frac{1}{3}\right)^n = [as \, n \rightarrow \infty, \left(\frac{1}{3}\right)^n \rightarrow 0] = 0
 \end{aligned}$$

1.2.5 4.5

$$\sum_0^{\infty} e^{2n \ln \sin\left(\frac{\pi}{3}\right)}$$

$$\begin{aligned}
 & 4.5 \\
 & \sum_0^{\infty} e^{2n \ln \sin \frac{\pi}{3}} \\
 & e^{2n \ln \sin \frac{\pi}{3}} = e^{2n \ln \frac{\sqrt{3}}{2}} = \left[e^{2 \ln \frac{\sqrt{3}}{2}} \right]^n = \left[e^{\ln \frac{3}{4}} \right]^n = \left(\frac{3}{4} \right)^n \\
 & \text{I } a_n = \left(\frac{3}{4} \right)^{n-1} \\
 & \text{II } a_1 = 1 \quad \left. \vphantom{\sum} \right\} S_n = 1 \cdot \frac{1 - \left(\frac{3}{4} \right)^n}{1 - \frac{3}{4}} = \frac{1 - \left(\frac{3}{4} \right)^n}{\frac{1}{4}} = 4 \left[1 - \left(\frac{3}{4} \right)^n \right]
 \end{aligned}$$

$$\text{III } \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} 4 \left[1 - \left(\frac{3}{4} \right)^n \right] = \left[\text{as } n \rightarrow \infty, \left(\frac{3}{4} \right)^n \rightarrow 0 \right] = 4$$

$$\text{IV } R_n = S - S_n = 4 - 4 \left[1 - \left(\frac{3}{4} \right)^n \right] = 4 \left(\frac{3}{4} \right)^n$$

$$\text{V } \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} 4 \left(\frac{3}{4} \right)^n = \left[\text{as } n \rightarrow \infty, \left(\frac{3}{4} \right)^n \rightarrow 0 \right] = 0$$

1.2.6 4.6

$$\sum_1^{\infty} \frac{1}{n(n+1)}$$

h. 6

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \left[\frac{1}{n} - \frac{1}{n+1} \right]$$

$$\left. \begin{array}{l} a_1 = 1 - \frac{1}{2} = \frac{1}{2} \\ a_2 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \\ a_3 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \\ a_n = \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)} \\ a_5 = \frac{1}{5} - \frac{1}{6} = \frac{1}{30} \end{array} \right\} \begin{array}{l} a_n = \frac{1}{n(n+1)} \\ \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n(n+1)} = 0 \end{array}$$

$$\text{II} \quad \sum_{n=1}^{\infty} \left[\frac{1}{n} - \frac{1}{n+1} \right] = 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{n} - \frac{1}{n+1} = 1 - \frac{1}{n+1} \quad \text{as } n \rightarrow \infty$$

$$S_n = 1 - \frac{1}{n+1}$$

$$\text{III} \quad \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[1 - \frac{1}{n+1} \right] = \left[\text{as } n \rightarrow \infty, \frac{1}{n+1} \rightarrow 0 \right] = 1$$

$$\text{IV} \quad R_n = S - S_n = 1 - 1 + \frac{1}{n+1} = \frac{1}{n+1}$$

$$\text{V} \quad \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0$$

1.2.7 4.7

$$\frac{3}{1 \cdot 2} - \frac{5}{2 \cdot 3} + \frac{7}{3 \cdot 4} - \frac{9}{4 \cdot 5} + \dots$$

h. 7

$$\frac{3}{1 \cdot 2} - \frac{5}{2 \cdot 3} + \frac{7}{3 \cdot 4} - \frac{9}{4 \cdot 5} + \dots =$$

$$\text{I } a_n = \frac{(-1)^{n-1} \cdot (2n+1)}{n(n+1)} \quad , \quad \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{(-1)^{n-1} \cdot (2n+1)}{n(n+1)} = 0$$

$$\text{II. } S_n = S_b - S_c$$

$$S_b = \frac{3}{1 \cdot 2} + \frac{7}{3 \cdot 4} + \frac{11}{5 \cdot 6} + \dots = \frac{3}{1} - \frac{3}{2} + \frac{7}{3} - \frac{7}{4} + \frac{11}{5} - \frac{11}{6} + \dots$$

$$S_c = \frac{5}{2} - \frac{5}{3} + \frac{9}{4} - \frac{9}{5} + \frac{13}{6} - \frac{13}{7} + \dots$$

$$S_n = 3 - \frac{3}{2} - \frac{5}{2} + \frac{7}{3} + \frac{5}{3} - \frac{7}{4} - \frac{9}{4} + \frac{11}{5} + \frac{9}{5} - \frac{11}{6} - \frac{13}{6} + \dots$$

$$S_n = 3 - \frac{8}{2} + \frac{12}{3} - \frac{16}{4} + \frac{20}{5} - \frac{24}{6} + \dots$$

$$S_n = 3 - 4 + 4 - 4 + 4 - 4 + 4 - 4 + \dots + \frac{(-1)^{n-1} \cdot (2n+1)}{n(n+1)}$$

$$S_n = 3 + (-1)^{n-1} \cdot \frac{(2n+1)}{n(n+1)}$$

$$\text{III } \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[3 + (-1)^n \frac{2n+1}{n(n+1)} \right] = \left[\text{as } n \rightarrow \infty, \frac{2n+1}{n(n+1)} \rightarrow 0 \right] = 3$$

$$\text{IV } R_n = S - S_n = 3 - 3 + \frac{(-1)^n (2n+1)}{n(n+1)} = \frac{(-1)^n (2n+1)}{n(n+1)}$$

$$\text{V } \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{(-1)^n (2n+1)}{n(n+1)} = \left[\text{as } n \rightarrow \infty, \frac{(2n+1)}{n(n+1)} \rightarrow 0 \right] = 0$$

1.3 Problems, Section 5

Use the preliminary test to decide whether the following series are divergent or require further testing. Careful: Do not say that a series is convergent; the preliminary test cannot decide this.

1.3.1 5.1

$$\frac{1}{2} - \frac{4}{5} + \frac{9}{10} - \frac{16}{17} + \frac{25}{26} - \frac{35}{37} + \dots$$

5.1

$$\left. \begin{array}{l} \frac{1}{2} - \frac{4}{5} + \frac{9}{10} - \frac{16}{17} + \frac{25}{26} - \frac{36}{37} + \dots \\ \frac{1}{2} - \frac{2^2}{2^2+1} + \frac{3^2}{3^2+1} - \frac{4^2}{4^2+1} + \dots \end{array} \right\} \Rightarrow \sum_0^{\infty} \frac{n^2}{n^2+1} \Rightarrow a_n = \frac{n^2}{n^2+1}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n^2}{n^2+1} = \left[\frac{\infty}{\infty} \right] = \lim_{n \rightarrow \infty} \frac{n^2 \cdot 1}{n^2 \left(1 + \frac{1}{n^2} \right)} = 1$$

Ans: Divergent

1.3.2 5.2

$$\sqrt{2} + \frac{\sqrt{3}}{2} + \frac{\sqrt{4}}{3} + \frac{\sqrt{5}}{4} + \frac{\sqrt{6}}{5} + \dots$$

5.2

$$\left. \begin{array}{l} \sqrt{2} + \frac{\sqrt{3}}{2} + \frac{\sqrt{4}}{3} + \frac{\sqrt{5}}{4} + \frac{\sqrt{6}}{5} + \dots \\ \frac{(1+1)^{\frac{1}{2}}}{1} + \frac{(1+2)^{\frac{1}{2}}}{2} + \frac{(1+3)^{\frac{1}{2}}}{3} + \frac{(1+4)^{\frac{1}{2}}}{4} + \dots \end{array} \right\} \Rightarrow \sum_1^{\infty} \frac{(n+1)^{\frac{1}{2}}}{n} \Rightarrow a_n = \frac{(n+1)^{\frac{1}{2}}}{n}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{(n+1)^{\frac{1}{2}}}{n} = \left[\frac{\infty}{\infty} \right] = \lim_{n \rightarrow \infty} \frac{\frac{1}{2}(n+1)^{-\frac{1}{2}} \cdot (n+1)'}{1} =$$

$$= \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1}} = \infty, \text{ Ans: } \underline{\text{Test further}}$$

1.3.3 5.3

$$\sum_{n=1}^{\infty} \frac{n+3}{n^2+10n}$$

5.3

$$\sum_{n=1}^{\infty} \frac{n+3}{n^2+10n} \Rightarrow a_n = \frac{n+3}{n^2+10n},$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n+3}{n^2+10n} = \lim_{n \rightarrow \infty} \frac{n^2 \left(\frac{1}{n} + \frac{3}{n^2} \right)}{n^2 \left(1 + \frac{10}{n} \right)} = 0$$

Ans: Test further

1.3.4 5.4

$$\sum_{n=1}^{\infty} \frac{(-1)^n n^2}{(n+1)^2}$$

5.4

$$\sum_{n=1}^{\infty} \frac{(-1)^n \cdot n^2}{(n+1)^2} \Rightarrow a_n = \frac{(-1)^n \cdot n^2}{(n+1)^2}, \text{ Suppose } f(x) = \frac{(-1)^x \cdot x^2}{(x+1)^2}$$

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{(-1)^x \cdot x^2}{(x+1)^2} = \lim_{x \rightarrow \infty} \frac{x^2 (-1)^x}{x^2 (1 + \frac{2}{x} + \frac{1}{x^2})} \rightarrow \text{doesn't exist}$$

Answer: The limit doesn't exist, so the series is divergent.

1.3.5 5.5

$$\sum_{n=1}^{\infty} \frac{n!}{n!+1}$$

5.5

$$\sum_{n=1}^{\infty} \frac{n!}{n!+1} \Rightarrow a_n = \frac{n!}{n!+1} = \frac{1 \cdot 2 \cdot 3 \cdots n}{1 + 1 \cdot 2 \cdot 3 \cdots n}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n!}{n!+1} = 1 \quad \text{It tends to 1, because } n! \text{ values grows far beyond 1}$$

Answer: Divergent

1.3.6 5.6

$$\sum_{n=1}^{\infty} \frac{n!}{(n+1)!}$$

5.6

$$\sum_{n=1}^{\infty} \frac{n!}{(n+1)!} \Rightarrow a_n = \frac{n!}{(n+1)!}, \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n!}{(n+1)!} = 0$$

It is because, at some point, denominator grows much faster than numerator does.

Ans: Test further

1.3.7 5.7

$$\sum_{n=1}^{\infty} \frac{(-1)^n n}{\sqrt{n^3 + 1}}$$

5.7

$$\sum_{n=1}^{\infty} \frac{(-1)^n n}{\sqrt{n^3 + 1}} \Rightarrow a_n = \frac{(-1)^n n}{\sqrt{n^3 + 1}}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{(-1)^n n}{\sqrt{n^3 + 1}} = \lim_{n \rightarrow \infty} \frac{n \cdot (-1)^n}{n \sqrt{n + \frac{1}{n^2}}} = 0$$

Ans: Test further

1.3.8 5.8

$$\sum_{n=1}^{\infty} \frac{\ln n}{n}$$

5.8

$$\sum_{n=1}^{\infty} \frac{\ln n}{n} \Rightarrow a_n = \frac{\ln n}{n}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{\ln n}{n} \xrightarrow{n=e^t} \lim_{t \rightarrow \infty} \frac{\ln e^t}{e^t} = \lim_{t \rightarrow \infty} \frac{t}{e^t} = 0$$

Answer: Test further

1.3.9 5.9

$$\sum_{n=1}^{\infty} \frac{3^n}{2^n + 3^n}$$

5.9

$$\sum_{n=1}^{\infty} \frac{3^n}{2^n + 3^n} \Rightarrow a_n = \frac{3^n}{2^n + 3^n}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{3^n}{3^n + 2^n} = \lim_{n \rightarrow \infty} \frac{3^n \cdot 1}{3^n(1 + \frac{2^n}{3^n})} = 1$$

$\downarrow 0$

Answer: Divergent

1.3.10 5.10

$$\sum_{n=2}^{\infty} \left(1 - \frac{1}{n^2}\right)$$

5.10

$$\sum_{n=2}^{\infty} \left(1 - \frac{1}{n^2}\right) \Rightarrow a_n = 1 - \frac{1}{n^2}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left[1 - \frac{1}{n^2}\right] = \left[\text{as } n \rightarrow \infty, \frac{1}{n^2} \text{ tends to } 0\right] = 1$$

Answer: Divergent

1.3.11 5.11

Give a proof of the preliminary test. Hint: $S_n - S_{n-1} = a_n$

5.11

Preliminary test proof: Suppose a convergent series

$$\lim_{n \rightarrow \infty} S_n = S \Rightarrow \lim_{n \rightarrow \infty} S_{n-1} = S$$

$$S_n - S_{n-1} = a_n$$

$$\lim_{n \rightarrow \infty} [S_n - S_{n-1}] = \lim_{n \rightarrow \infty} a_n$$

$$\lim_{n \rightarrow \infty} a_n = S - S = 0 \quad \blacksquare$$

1.4 Problems, Section 6

[]:

1.4.1 6.5 b)

Test the following series for convergence using the comparison test.

$$\sum_{n=2}^{\infty} \frac{1}{\ln n}$$

Proof of Divergence for $H = \sum \frac{1}{\ln n}$ To determine the convergence of the series $\sum_{n=2}^{\infty} \frac{1}{\ln n}$, we will use the Direct Comparison Test.

We choose the harmonic series as our reference divergent series:

$$D = \sum_{n=2}^{\infty} \frac{1}{n}$$

(We know from previous proofs that the harmonic series is divergent).

Now, we compare the terms of both series. For all $n \geq 2$, it is a known property of the natural logarithm that it grows slower than linear function:

$$\ln n < n$$

Since the denominator of our H series is strictly smaller than the denominator of the D series, taking the reciprocal reverses the inequality:

$$\frac{1}{\ln n} > \frac{1}{n}$$

Since our series H is strictly greater term-by-term than the divergent harmonic series D , it follows by the Direct Comparison Test that H must also be divergent:

$$\sum_{n=2}^{\infty} \frac{1}{\ln n} = \infty$$

```
[5]: def plot_graph():
    n_max = 50
    n_start = 2
    n_step = 1
    n_values = np.arange(n_start, n_max, n_step)
    h_values = 1/ np.log(n_values)
    c_values = 1/ 2 ** n_values
    d_values = 1/ n_values

    plt.figure(figsize=(16,9))

    plt.scatter(n_values, h_values, color='red', marker='o', s=40, label=r'$H$')
    ↪series: $\frac{1}{\ln n}$')
    plt.scatter(n_values, c_values, color='blue', s=40, marker='^', label=r'$C$')
    ↪series: $\frac{1}{2^n}$')
    plt.scatter(n_values, d_values, color='green', s=40, marker='*',
    ↪label=r'$D$ series: $\frac{1}{n}$')

    plt.legend(loc='upper right', fontsize=20)
    plt.title(r"The graph of serieses $H$, $C$ and $D$", fontsize=20, pad= 20)

    plt.show()

    table_data = {
```

```

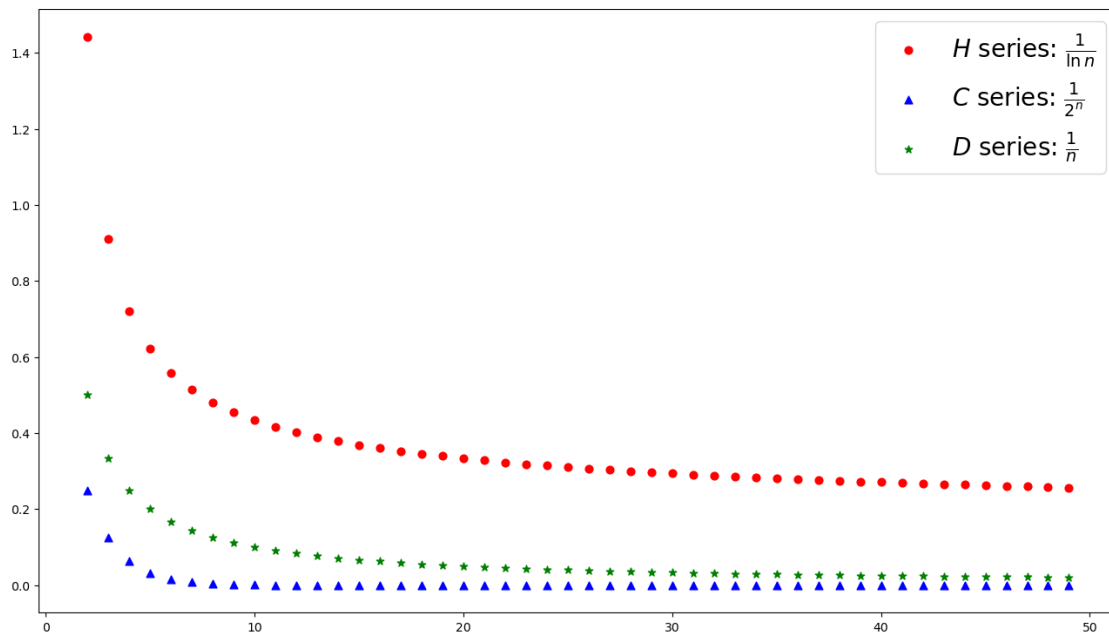
    r"Step $n$: n_values,
    r"Series $H$ values": np.round(h_values, 4) ,
    r"Series $C$ values": np.round(c_values, 4),
    r"Series $D$ values": np.round(d_values,4)
}

df = pd.DataFrame(table_data)
df_sliced = df.iloc[:5]
return Markdown(df_sliced.to_markdown(index=False))

plot_graph()

```

The graph of serieses H , C and D



[5]:

Step n	Series H values	Series C values	Series D values
2	1.4427	0.25	0.5
7	0.5139	0.0078	0.1429
12	0.4024	0.0002	0.0833
17	0.353	0	0.0588
22	0.3235	0	0.0455
27	0.3034	0	0.037
32	0.2885	0	0.0312
37	0.2769	0	0.027
42	0.2675	0	0.0238
47	0.2597	0	0.0213

$$D = \left[\frac{1}{10} + \dots (9 \text{ terms each} = \frac{1}{10}) \right] + \left[90 \text{ terms each} = \frac{1}{100} \right] + \dots$$